

1.2 IMPLICATIONS

Objectives

After completing this section, you should be able to do the following.

1. Explain the conditions under which an implication is true.
2. Identify statements as equivalent to a given implication or its converse.
3. Explain the relationship between the truth values of an implication, its converse, and its contrapositive.

1.2.1 SECTION PREVIEW

Investigate!

Little Timmy's Mom tells him, "If you don't eat all your broccoli, then you will not get any ice cream." Of course, Timmy loves his ice cream, so he quickly eats all his broccoli (which actually tastes pretty good).

After dinner, when Timmy asks for his ice cream, he is told no! Does Timmy have a right to be upset? Why or why not?

By far, the most important type of statement in mathematics is the implication. It is also the least intuitive of our basic molecular statement types. Our goal in this section is to become more familiar with this key concept.

To see why this sort of statement is so prevalent, consider the *Pythagorean Theorem*. Despite what social media might claim, the Pythagorean Theorem is not

$$a^2 + b^2 = c^2.$$

Okay, sure, that has a variable in it, so we must be using the convention to take the universal generalization,

$$\forall a, b, c \in \mathbb{R} (a^2 + b^2 = c^2).$$

So $1^2 + 5^2 = 2^2$??? Okay, fine. The equation is true as long as a and b are the lengths of the legs of a right triangle and c is the length of the hypotenuse. In other words:

If a and b are the lengths of the legs of a right triangle with hypotenuse of length c , then $a^2 + b^2 = c^2$.

Math is about making general claims, but a claim is rarely going to be true of absolutely *every* mathematical object. The way we *restrict* our claims to a particular type of object is with an implication: "Take any object you like, *if* it is of the right type, *then* this thing is true about it."

Similarly, as we saw in the Quantifiers and Predicates subsection, when we make claims like “Every square is a rectangle,” we really have an implication: “If something is a square, then it is a rectangle.”

Here is a reminder of what we mean by an implication.

Definition 1.2.1 Implication.

An **implication** (or **conditional**) is a molecular statement of the form

$$P \rightarrow Q$$

where P and Q are statements. We say that

- P is the **hypothesis** (or **antecedent**).
- Q is the **conclusion** (or **consequent**).

An implication is *true* provided P is false or Q is true (or both), and *false* otherwise. In particular, the only way for $P \rightarrow Q$ to be false is for P to be true and Q to be false.

The definition of truth of an implication can also be represented as a truth table:

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Figure 1.2.2 The truth table for $P \rightarrow Q$.

Does this truth table make sense? Should we believe it? Look in particular at the third row: F, T, T, and consider the implication, “If $5 < 3$ then $5 + 3 = 8$.” Does that statement *feel* true? The truth table says it should be (since $5 < 3$ is false, and $5 + 3 = 8$ is true).

Much of what we will do in the remainder of this section is convince ourselves that this truth table makes sense.

PREVIEW ACTIVITY

- Consider the statement, “If Tommy doesn’t eat his broccoli, then he will not get any ice cream.” Which of the following statements mean the same thing (i.e., will be true in the same situations)? Select all that apply.
 - If Tommy does eat his broccoli, then he will get ice cream.
 - If Tommy gets ice cream, then he ate his broccoli.
 - If Tommy doesn’t get ice cream, then he didn’t eat his broccoli.

- D. Tommy ate his broccoli and still didn't get any ice cream.
2. Suppose that your shady uncle offers you the following deal: If you loan him your car, then he will bring you tacos. In which of the following situations would it be fair to say that your uncle is a liar (i.e., that his statement was false)? Select all that apply.
- A. You loan him your car. He brings you tacos.
 - B. You loan him your car. He never buys you tacos.
 - C. You don't loan him your car. He still brings you tacos.
 - D. You don't loan him your car. He never brings you tacos.
3. Consider the *sentence*, "If $x \geq 10$, then $x^2 \geq 25$." This sentence becomes a statement when we replace x by a value, or "capture" the x in the scope of a quantifier. Which of the following claims are true (select all that apply)?
- A. If we replace x by 15, then the resulting statement is true. (Note, $15^2 = 225$.)
 - B. If we replace x by 3, then the resulting statement is true.
 - C. If we replace x by 6, then the resulting statement is true.
 - D. The universal generalization ("for all x , if $x \geq 10$ the $x^2 \geq 25$ ") is true.
 - E. There is a number we could replace x with that makes the statement false.
4. Consider the statement, "If I see a movie, then I eat popcorn" (which happens to be true). Based solely on your intuition of English, which of the following statements mean the same thing? Select all that apply.
- A. If I eat popcorn, then I see a movie.
 - B. If I don't eat popcorn, then I don't see a movie.
 - C. It is necessary that I eat popcorn when I see a movie.
 - D. To see a movie, it is sufficient for me to eat popcorn.
 - E. I only watch a movie if I eat popcorn.

1.2.2 UNDERSTANDING THE TRUTH TABLE

The truth value of the implication is determined by the truth values of its two parts. Our definition of the truth conditions for an implication says that there is only one way for an implication to be false: when the hypothesis is true and the conclusion is false.

Example 1.2.3

Consider the statement:

If Bob gets a 90 on the final, then Bob will pass the class.

This is definitely an implication: P is the statement “Bob gets a 90 on the final,” and Q is the statement “Bob will pass the class.”

Suppose I made that statement to Bob. In what circumstances would it be fair to call me a liar? What if Bob really did get a 90 on the final, and he did pass the class? Then I have not lied; my statement is true. However, if Bob did get a 90 on the final and did not pass the class, then I lied, making the statement false. The tricky case is this: What if Bob did not get a 90 on the final? Maybe he passes the class, maybe he doesn’t. Did I lie in either case? I think not. In these last two cases, P was false, and the statement $P \rightarrow Q$ was true. In the first case, Q was true, and so was $P \rightarrow Q$. So $P \rightarrow Q$ is true when either P is false or Q is true.

Just to be clear, although we sometimes read $P \rightarrow Q$ as “ P implies Q ”, we are not insisting that there is some *causal* relationship between the statements P and Q (although there might be). “If $x < y$, then $x + 1 < y + 1$,” is a true statement (or at least, its universal generalization is). We know it is true because we understand how the two parts interact. If you add 1 to two numbers x and y , then their order does not change. But the statement, “if $1 < 2$, then Euclid studied geometry” is also a true implication.

Example 1.2.4

Decide which of the following statements are true and which are false. Briefly explain.

1. If $1 = 1$, then most horses have 4 legs.
2. If $0 = 1$, then $1 = 1$.
3. If 8 is a prime number, then the 7624th digit of π is an 8.
4. If the 7624th digit of π is an 8, then $2 + 2 = 4$.

Solution. All four of the statements are true. Remember, the only way for an implication to be false is for the *if* part to be true and the *then* part to be false.

1. Here both the hypothesis and the conclusion are true, so the implication is true. It does not matter that there is no meaningful connection between the true mathematical fact and the fact about horses.

2. Here the hypothesis is false and the conclusion is true, so the implication is true.
3. I have no idea what the 7624th digit of π is, but this does not matter. Since the hypothesis is false, the implication is automatically true.
4. Regardless of the truth value of the hypothesis, the conclusion is true, making the implication true.

This is a strange example and isn't really how we use implications anyway. This strangeness is not just mathematicians being stubborn though. The truth conditions for implications *must* be like they are for mathematics to make sense. Let's see why.

Example 1.2.5

Consider the statement, "All squares are rectangles," which can also be phrased as, "For all shapes, if the shape is a square, then it is a rectangle." Is this statement true or false? Are we sure? What about the following three shapes?



Solution. Of course the statement is true. A square is a 4-sided plane figure with 4 right angles and 4 equal-length sides, while a rectangle is a 4-sided plane figure with 4 right angles.

However, what we mean when we consider a universal statement like this is that, no matter what we "plug in" for the variable ("the shape" in this case), the resulting statement is true. When the statement is about a particular shape, we have an implication $P \rightarrow Q$. This means it must be true that, if the actual shape on the left is a square, then it is a rectangle. Great. The shape is a square (P is true) and is a rectangle (Q is true), so yes, the implication is true.

Is the implication true of the rectangle in the middle? Well, that shape is not a square (P is false), and it is a rectangle (Q is true). But look, we believe that all squares are rectangles, so the statement must be true. Even of a rectangle. The only way this works is if "false implies true" is true!

Similarly, all squares are rectangles is a true statement, even when we look at a triangle. P is false (the triangle is not a square), and Q is false (the triangle is not a rectangle). Thankfully, we defined implications to be true in this case as well.

We have given shapes that illustrate lines 1, 3, and 4 of the truth table for implications (Figure 1.2.2). What shape illustrates line 2? That would need to